Substance Use & School Violence Inference Analysis

RESEARCH QUESTION

Does the frequency of adolescent substance use (smoking & alcohol) significantly predict the risk of school violence (fighting & carrying weapons)?

METHOD

- Ordinary Least Squares (OLS) Multiple Linear Regression
- Violence = β0+ β1·Cig + β2·Alc +β3·Age + β4·Sex + ε

RESULTS

Predictor	Coef (β)	p-value
Smoking Days	0.0260	<0.001
Drinking Days	0.0849	<0.001
Gender (Male)	0.5093	<0.001
Age	-0.0736	<0.001

R-squared: 0.050 | F-statistic: 156.8 (p < 0.001)

DATA & VARIABLES

- Dependent (Y): School Violence Index (Fights + Weapons)
- Independent (X): Cigarette Smoking Days,Alcohol Drinking Days
- Controls: Age, Gender

EDA

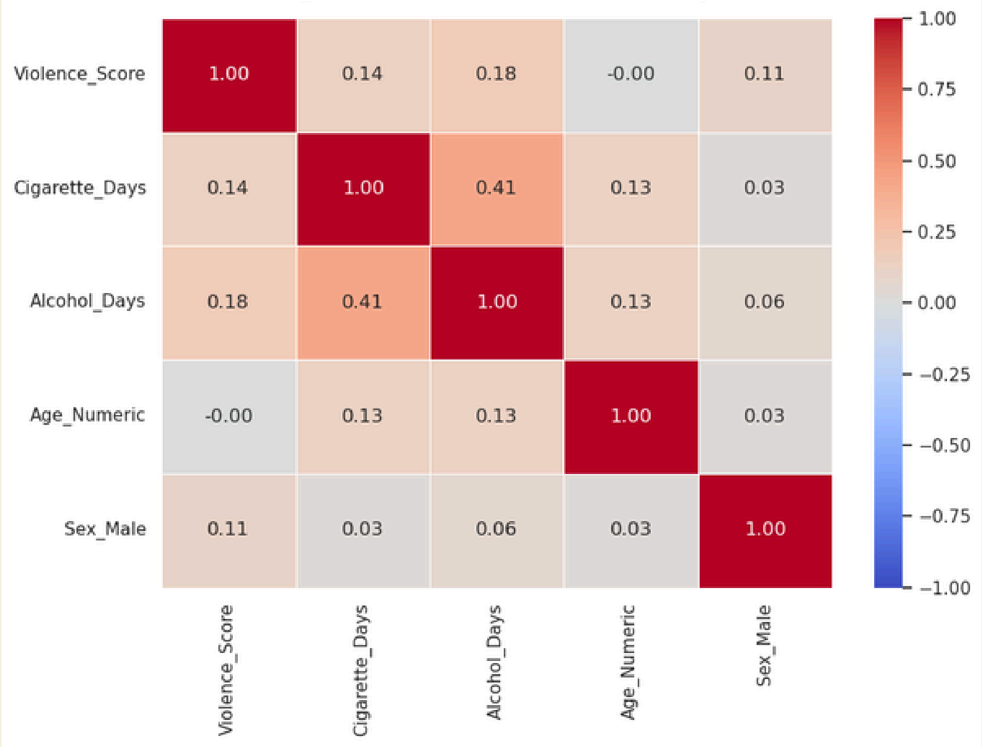

Figure 1: Correlation Matrix Heatmap

As shown in the heatmap, based on a massive sample size of N= 11,904, we see a strong positive correlation cluster between substance use and school violence risk.

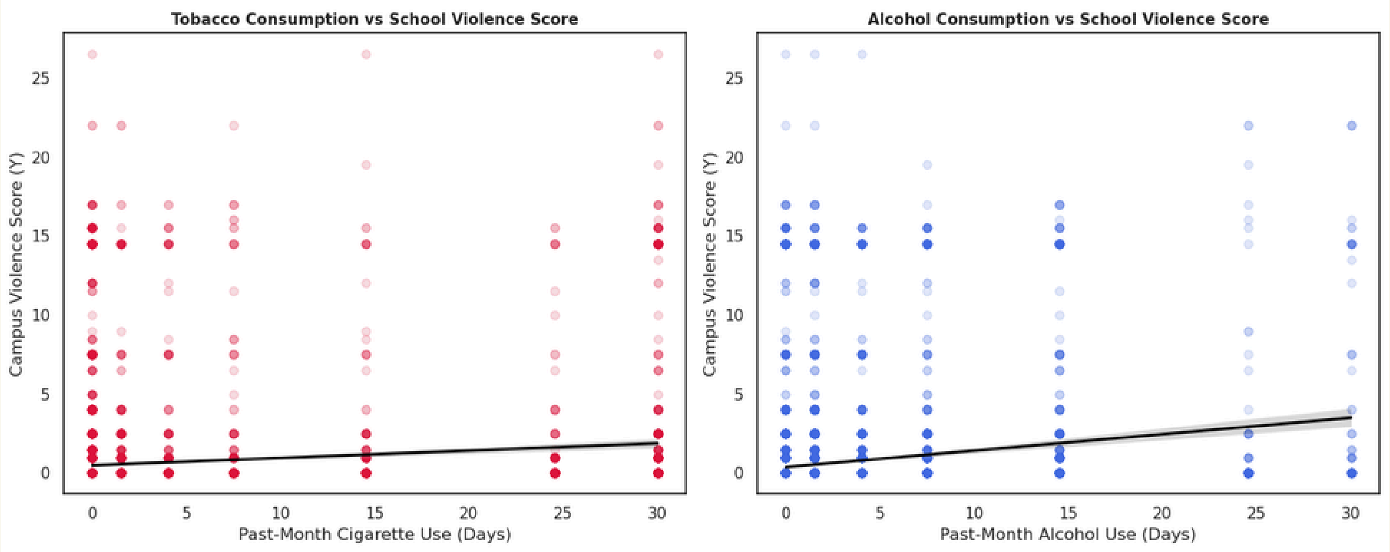

Figure 2: Linear Regression Trends for Substance Abuse Predictors

Illustrates that as the continuous days of cigarette or alcohol consumption scale upward, the baseline baseline threshold for student involvement in school property physical altercations and weapon carriage expands linearly.

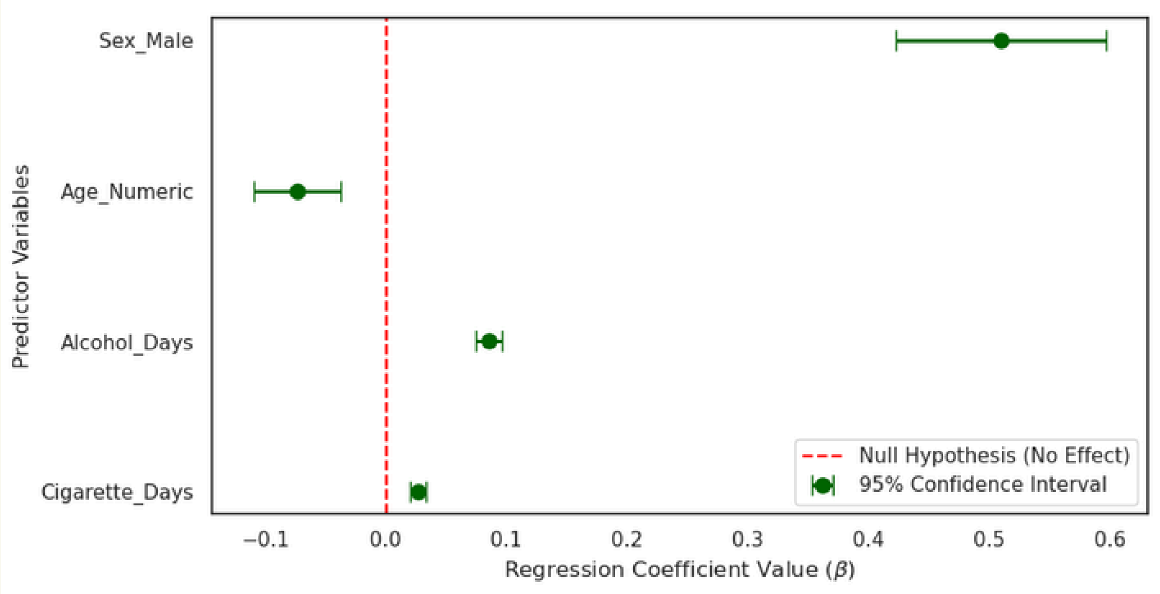

Figure 3: Predictor Coefficients with 95% Confidence Intervals

The coefficient plot confirms that smoking, drinking, and male status are significant positive predictors, with all 95% confidence intervals above zero and p < 0.001.

(Conclusion)

The null hypothesis is successfully rejected. Empirical evidence confirms that the frequency of adolescent substance use is a robust, positive predictor of school violence risks.